ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ROBOTICS FUNDAMENTALS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Robotics Fundamentals paper 1 · practice a.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Accept answers in the student's own words if the meaning is correct, and other correct examples. Use consistent accuracy in calculations.

SECTION A

QUESTION 1

1.1 1.1.1 D ✓ 1.1.2 B ✓ 1.1.3 D ✓ 1.1.4 B ✓ 1.1.5 C ✓ 1.1.6 A ✓ 1.1.7 B ✓ 1.1.8 D ✓ 1.1.9 A ✓ 1.1.10 C ✓ (10)

1.2 1.2.1 False ✓ 1.2.2 True ✓ 1.2.3 True ✓ 1.2.4 True ✓ 1.2.5 False ✓ 1.2.6 True ✓ 1.2.7 True ✓ 1.2.8 True ✓ 1.2.9 False ✓ 1.2.10 True ✓ (10)

1.3 1.3.1 A ✓ 1.3.2 C ✓ 1.3.3 B ✓ 1.3.4 D ✓ 1.3.5 J ✓ 1.3.6 I ✓ 1.3.7 G ✓ 1.3.8 F ✓ 1.3.9 H ✓ 1.3.10 E ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 A robot is reprogrammable and can do different tasks ✓; it uses sensors to respond to its environment and make decisions ✓; an ordinary machine does one fixed task in the same way every time (e.g. a washing machine runs set cycles) ✓. (3)

2.2 Any FIVE industries with a task (1 + 1 each): motor manufacturing – welding and painting car bodies ✓✓; mining – inspection of shafts, rescue robots, drilling ✓✓; agriculture – fruit picking, drones spraying crops ✓✓; medicine – surgical robots ✓✓; food and beverages – packing, bottling ✓✓; warehousing – moving goods; security – bomb disposal; electricity (Eskom) – drones inspecting power lines (10)

2.3 Advantages (any FOUR) ✓✓✓✓: work 24 hours without rest; consistent quality / accuracy; do dangerous work (painting fumes, welding); faster production; lower long-term labour costs. Disadvantages (any THREE) ✓✓✓: high cost to buy and install; job losses; need skilled technicians for programming and maintenance; cannot adapt to unexpected situations like people; breakdowns stop production (7)

[20]

QUESTION 3

3.1 Name ✓ + function ✓ each: controller – runs the program and makes decisions; sensors – gather information from the environment; actuators (motors) – produce movement; power supply – provides energy (battery, mains, compressed air); end effector / manipulator – performs the task (gripper, tool); (accept body / chassis – holds the parts together) (10)

3.2 Infrared (IR) line sensor / LDR ✓ – shines IR light on the floor; white reflects a lot of light and black reflects little, so the robot can tell where the line is ✓✓; ultrasonic sensor ✓ – sends out a sound pulse and measures the time for the echo to return to calculate the distance to an object ✓✓ (6)

3.3 DC motor – driving the wheels (continuous rotation) ✓; servo motor – moving a gripper or steering to an exact angle ✓; stepper motor – precise positioning in steps, e.g. 3D printer or robot axis ✓ (3)

3.4 Open loop: no feedback – the robot carries out commands without checking the result ✓, e.g. drives forward for 3 seconds whether or not it hits something ✓; closed loop: sensors feed information back to the controller, which corrects the action ✓, e.g. an encoder checks that the wheels turned the correct distance / the robot stops when the ultrasonic sensor detects a wall ✓. (4)

[23]

QUESTION 4

4.1 Any THREE configurations with uses (1 + 1 each): Cartesian (gantry) – pick and place, CNC, 3D printing ✓✓; cylindrical – assembly, handling machine tools ✓✓; spherical (polar) – spot welding, die casting ✓✓; SCARA – fast assembly of electronics; articulated – welding, painting car bodies (6)

4.2 Shoulder – base / waist and shoulder joint ✓; elbow – elbow joint ✓; wrist – wrist joint (pitch, yaw, roll) ✓; hand – end effector / gripper ✓ (4)

4.3 $\text{Circumference} = \pi \times \text{d} = 3{,}14 \times 6{,}5 \checkmark = 20{,}41 \text{ cm} \checkmark; \text{distance} = 10 \times 20{,}41 = 204{,}1 \text{ cm} (\approx 2{,}04 \text{ m}) \checkmark$ (3)

4.4 Any TWO: legs (walking robots) ✓; tracks (like a tank) ✓; propellers (drones, underwater robots); snake-like crawling; rolling (spherical robots) (2)

[15]

QUESTION 5

5.1 Any FOUR: safety fences / cages around the robot ✓; light curtains or pressure mats that stop the robot when a person enters ✓; emergency stop buttons ✓; lock-out/tag-out during maintenance ✓; warning lights and signs; training of workers; reduced speed mode when people are near; PPE (4)

5.2 For (any TWO) ✓✓: robots replace workers in repetitive jobs (packing, assembly); companies save money by using fewer workers. Against (any TWO) ✓✓: robots create new jobs (programmers, technicians); dangerous jobs are taken over so workers are safer; productivity grows and businesses expand; people move to higher-skilled work with training (4)

5.3 Any FOUR: robot programmer ✓; robotics / mechatronics technician ✓; maintenance technician ✓; drone pilot ✓; automation engineer; robot sales and support; robotics teacher / trainer; AI and machine-vision developer (4)

[12]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE